TEXORA · LECTURE SHEET

Single Jersey Structure

The plain weft-knitted structure taken as a piece of engineering geometry: stitch length as the master variable, tightness factor as the compactness scale, the GSM identity that ties counts to weight, and the characteristic behaviours — curling, spirality, laddering — that follow from knitting every loop in the same direction.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 10 · LEVEL 3 TERM 2

1. The structure restated

Single jersey is knitted on **one needle bed** with **every loop drawn in the same direction** (notation: every course ● ● ● ●, Topic 9). The technical face shows the V legs of the needle loops; the technical back shows the head arcs and sinker loops. It is the lightest, cheapest and most productive weft structure — and the least balanced, because all the bending energy of the yarn points the same way. Everything in this sheet is a consequence of that single sentence.

2. Stitch length — the master variable

The **stitch length** *l* is the length of yarn in one complete stitch (needle loop + sinker loop), measured in mm on yarn unravelled from relaxed fabric (the TE-3102 method: length of a course ÷ number of wales in it). Stitch length is set on the machine by the cam (stitch) setting and yarn input tension, and it is remarkably robust: in the relaxed fabric, *l* — not the machine's instantaneous CPC or WPC — is the quantity that survives handling, washing and relaxation unchanged. Hence the working doctrine of knitted structure: **specify the stitch length, and the relaxed geometry follows**.

3. Tightness factor

Compactness across different yarn counts is compared by the **tightness factor**

$$K = \sqrt{(\text{yarn count in tex})} \div \text{stitch length l (mm)}$$

WORKED EXAMPLE — TIGHTNESS FACTOR

A cotton single jersey is knitted from 19.7 tex yarn at a stitch length of 3.1 mm.

K = √19.7 ÷ 3.1 = 4.44 ÷ 3.1 = **1.43**.

For cotton single jersey, K ≈ 1.3 is slack (loose, soft, prone to distortion), K ≈ 1.4–1.5 is the

normal commercial range, and K ≈ 1.6 is tight (firm, heavy for its count, harder to knit without faults). The sample of TE-3206/2 (19.7 tex, l = 3.0 mm) gives K = 4.44 ÷ 3.0 = **1.48** — at the firm end of normal, which its measured geometry (Topic 10, Section 4) confirms.

Because K combines count and stitch length, two fabrics of different counts but equal K are geometrically similar — loops of the same shape at different scales — which is why K, not stitch length alone, is the number quoted when a quality is re-engineered in a new count.

4. The GSM identity and relaxed geometry

One square metre of relaxed fabric contains (WPC × 100) wales in the width and (CPC × 100) courses in the length; each stitch holds *l* mm of yarn of the given tex. Multiplying out gives the identity used throughout TE-3102/3206:

$$\text{GSM} = \text{WPC} \times \text{CPC} \times l\,(\text{mm}) \times \text{tex} \div 100$$

WORKED EXAMPLE — GSM FROM STRUCTURE

Single jersey, relaxed state: WPC = 12 per cm, CPC = 18 per cm, l = 3.1 mm, yarn 19.7 tex.
GSM = 12 × 18 × 3.1 × 19.7 ÷ 100 = 216 × 3.1 × 19.7 ÷ 100 = 13,191 ÷ 100 ≈ **132 g/m²**.
The same identity run backwards is the production engineer's tool: a target GSM with a chosen count fixes the required WPC × CPC × l product, and K then chooses how the product is shared between stitch length and density.

Relaxation states must ride with every such number: **dry-relaxed** (off the machine, conditioned), **wet-relaxed** (after a soak and flat drying), **fully relaxed/finished** (after the fabric's actual finishing route). WPC and CPC both rise as the fabric relaxes (the loops re-shape and the fabric contracts), while *l* and hence the identity keep the books balanced — a GSM change between states is a geometry change, not a yarn change.

Width from needles. The wales of a tubular jersey are the machine's needles: relaxed open width = total needles ÷ WPC. A 1,671-needle machine (19-inch, 28 gauge — the machine estimated in TE-3206/2) at WPC 12 per cm gives 1,671 ÷ 12 ≈ **139 cm** open width in the relaxed state, from a tube circumference on the machine of π × 19 in ≈ 152 cm — the difference is the width contraction of relaxation, and it is why "width" without a state is not a specification.

5. Characteristic behaviours

- **Curling:** the unbalanced loop bending rolls the fabric — toward the face at the top and bottom edges, toward the back at the side edges. Managed by finishing (heat-setting for synthetics, resins,

compacting) and by garment construction (hems, bands of rib/interlock).

- **Spirality:** courses rest at a slight angle to the wales instead of square, twisting garments after washing. Its roots are the residual torque of the spun yarn (a lively singles yarn keeps trying to untwist, skewing each loop) feeding always in the same rotational sense on a single-bed machine. Controls: twist-balanced or plied yarns, heat-setting, alternating feeder yarn rotation (S/Z alternation), resin/compactor finishing. Measured as the angle or the percentage displacement of a course-end marker after washing — a number TE-3201's garments-testing syllabus meets again.
- **Laddering:** one broken loop releases its wale stitch by stitch; jersey ladders readily from either end of the fabric, as TE-3101 established.
- **Stretch and recovery:** large widthwise extension from loop re-shaping, modest lengthwise; recovery depends on tightness (higher K, better recovery) and on any elastane plated in.

6. Derivatives in one paragraph

Needle selection on the same single bed gives the jersey family: **stripes** (feeder yarn changes), **piqué and lacoste** (tuck/miss courses in repeat), **single-jersey fleece** (an inlay yarn floated on the back and brushed), **plated fabrics** (two yarns per stitch, one shown on each face), **jersey terry** (plated sinker-loop pile). Each is specified by the same five parameters plus its selection repeat — the basics of Topic 9 with a patterning instruction added.

Glossary

Stitch length — yarn length per stitch (needle loop + sinker loop), mm. **Tightness factor K** — √tex ÷ l (mm). **Relaxation state** — dry-, wet- or fully relaxed condition in which geometry is quoted. **Spirality** — skewing of courses relative to wales after relaxation/washing. **Plating** — feeding two yarns so each shows on a chosen face of the stitch.

Key points

• Stitch length is the master variable: it survives relaxation when WPC/CPC do not.
• K = √tex ÷ l: cotton jersey ≈1.3 slack, 1.4–1.5 normal, 1.6 tight (worked: 1.43; TE-3206 sample 1.48).
• GSM = WPC × CPC × l × tex ÷ 100 (worked: ≈132 g/m²); width = needles ÷ WPC (worked: ≈139 cm relaxed).

Practice

1. A jersey is knitted from 24.6 tex yarn at l = 3.4 mm. Compute K and classify the tightness.
2. From WPC 13, CPC 20, l = 3.0 mm, 16.4 tex, compute the GSM.

3. The same fabric relaxes further and its CPC rises from 18 to 20 with WPC unchanged. State what happens to its GSM and to its stitch length, with the reason.
4. Give the two independent controls a mill can apply against spirality before finishing, and the finishing control.

Answer key

1. K = √24.6 ÷ 3.4 = 4.96 ÷ 3.4 = **1.46** — normal commercial tightness for cotton jersey.
2. GSM = 13 × 20 × 3.0 × 16.4 ÷ 100 = 260 × 3.0 × 16.4 ÷ 100 = 12,792 ÷ 100 ≈ **128 g/m²**.
3. GSM rises in the ratio 20/18 (about +11%, to ≈147 g/m² by the identity): the same yarn is now packed into a smaller area because the fabric has contracted in length. The stitch length does **not** change — it is yarn per loop, and no yarn has been added or removed; only the loop shape changed. This is exactly why l, not CPC, is the state-independent variable.
4. Before finishing: (i) use twist-balanced yarns (plied yarns, or alternate S- and Z-twist yarns feeder by feeder so loop torques cancel); (ii) reduce yarn liveliness at source (steam-setting the yarn / choosing lower residual torque). Finishing control: heat-setting (synthetics) or resin/compacting treatment that locks the relaxed geometry (cotton).

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